AI BASED INTERACTIVE ROBOT FOR ELDERLY PEOPLE

A PROJECT REPORT

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ABSTRACT

AI has revolutionized the world and is becoming increasingly popular in our day-to-day life. With massive improvements in healthcare and lifestyle in the last century, people are living longer. While longevity is an important achievement of modern-day, it does present challenges in terms of caring for an increasingly elderly population. The issue of elder care is one of supply and demand. As the elder population increases in numbers, the number of caregivers is not correspondingly increasing. Japan predicts a shortage of 1 million caregivers by the year 2025. And the United States predicts that the percentage of people aged 65 and older is expected to increase to roughly 26% of the population by 2050. The relative lack of people able to care for the elderly is what keeps the costs of elderly care so high and creates a burden on family and caregivers. One solution to this supply and demand issue is the development and adoption of elder care robots. There is a growing market of elderly people every year and yet the number of elderly care robots are very low compared to other service segment robots. **"AI based interactive robot for elderly people"** is a robot designed with a built-in voice assistant with artificial intelligence, which interacts with the elderly people. This is done on a Jetson nano board which is a dedicated development board for AI and machine learning. This is also helpful in making a robot at low cost which can be affordable to all civilians.

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CHAPTER 1

INTRODUCTION

Motivation

The issue of elder care is one of supply and demand. As the elder population increases in numbers, the number of caregivers is not correspondingly increasing. Japan predicts a shortage of 1 million caregivers by the year 2025. And the United States predicts that the percentage of people aged 65 and older is expected to increase to roughly 26% of the population by 2050. The relative lack of people able to care for the elderly is what keeps the costs of elderly care so high and creates a burden on family and caregivers.

Objective

Using medical robots for elderly care will vastly reduce the current astronomical cost of elderly care. Medical robots are a growing industry. According to the International Federation of Robotics World Robotics 2018 Service Robots report, medical robot sales increased 73% in 2017 over 2016, accounting for 2.7% of all professional service robot sales.

There are numerous ways that medical robotics can help the elderly:

- Robots can perform small tasks like fetching food and water.
- Some elder care robots handle social and emotional needs by providing entertainment through games, helping remind them of events and appointments, and providing social engagement.
- Other eldercare robots take a more direct, muscular approach and use powerful hydraulics to help provide mobility and transportation support to seniors.

Approach

The fast adoption and rising performance of artificial intelligence drive interest in voice assistants. Voice assistants can understand voice commands, whether for answering a question, playing music, or turning on the lights. To talk naturally to people, assistants should use a combination of AI technologies:

- **Automatic speech recognition** (ASR) is a voice processing component which converts spoken language into textual representation.
- **Natural language understanding** (NLU) translates user queries from natural language into a formal semantic representation. This representation is then used by the system to select the next action and generate a response. Usually, NLU requires completion of a variety of tasks, such as text normalization, morphological analysis, semantic analysis, ranking and named entity recognition.
- **Text-to-speech** (TTS) takes the response in a text format and produces its voice representation.

Obstacle avoiding robot senses obstacles in the path, avoid it and resumes its running. There are some very popular methods for robot navigation like wall-following, edge detection, line following and many more. A more general and commonly employed method for obstacle avoidance is based on edge detection.

Result

Thus, we have built an AI based voice assistant along with Obstacle avoidance interfaced to the robot which can interact with elderly people and takes care of them.

CHAPTER 2

LITERATURE REVIEW

A large number of studies on the use of artificial intelligence for voice assistant and obstacle avoidance have been carried out in the past. For the purpose of review, research has been defined as papers published in e-journals and international conferences on voice assistance and obstacle avoidance using artificial intelligence and machine learning.

Brief reviews of the following research works reveal the wealth of contributions on voice assistance and obstacle avoidance using artificial intelligence.

1. In October 2020 Ryosuke Miyata, Yunan He, Osamu Fukuda Nobuhiko Yamaguchi and Hiroshi Okumura, Anik Nur Handayani developed an AI autonomous mobile robot based on open-source software. This robot is capable of voice recognition and object recognition, and with these functions, the robot can find and approach the object specified though a user's voice. Since the robot is controlled with the voice commands, the user can control the robot intuitively. They used a software platform named Robot Operating System (ROS) to speed up the development. All functions including voice recognition, object recognition, and robot control are implemented in a single board computer based on open-source software. They conducted the preliminary experiments and verified the performance of the developed system.

2. In January 2020, Zhiheng Wang, Perry Xiao, Daqing Chen proposed a study to design a 6-degree of freedom robotic arm which can pick up objects in random positions on a 2D surface based on Arduino microcontroller, ultrasonic sensors and picamera. The robotic arm is able to recognise objects based on computer vision algorithm for shape detection. The ultrasonic sensor measures the distance between

the objects and the robotic arm, and the position of the objects in the real world will be detected by its mass centre in the image to improve the accuracy of the pick-up movement. Arduino microcontroller will calculate the rotation angles for the joints of the robotic arm by using inverse kinematics algorithms.

3. In August 2020, Wuttichai Vijitkunsawat, Peerasak Chantngarm presented an approach that the vehicle is used in abundant activities, which lead to an increasing number of accidents on the roads. Many statistical reports pointed that more than 80% of accident causes came from direct human causes such as violating the speed limit, illegal overtaking, and suddenly cutting in. Therefore, the self-driving car was rapidly developed by starting a scaled RC-Car platform. This paper presents a self-driving car model to study behavior by using the three machine learning algorithms: SVM, ANN-MLP, and CNN-LSTM in 3-speed levels and 3 scenarios, both with obstacles and without obstacle. The results show that CNN-LSTM is the best accuracy in every scenario and speed levels.

4. In the year 2013, C. Park, S. Kang, J. Kim and J. Oh presented this paper on elder care robot system for older people. An elder care robot system provides human-robot interaction application and monitoring service. The study was carried out for five weeks in nursing homes, with the participation of 17 residents. They have conducted a t-test to compare the difference between satisfaction and usability evaluation for the residents and the staff.

5. In August 2016, M. Aman, M. Mahmud, H. Jiang, A. Abdelgawad and K. Yelamarthi proposed an obstacle detection and navigation of dynamic environments is a challenge in mobile robotics. To address this challenge, this paper presents an efficient sensor fusion methodology to detect the size and location of obstacles and navigate the mobile robot with high accuracy. This is done by leveraging upon the

unique advantages of accuracy in both ultrasonic sensor and a Kinect sensor for near-field and far-fields respectively. Further, an efficient Kalman filter is implemented to reduce the systematic errors in encoder data to track robot pose of the robot in real-time and reach the destination with high accuracy. Implemented on differential drive-based mobile robot, the proposed system has been validated with a high efficiency of detecting obstacles and reaching the destination with an accuracy of 5cm.

6. In Nov 2017, Chiao-Yu Yang, Ming-Jen Lu, Shih-Huan Tseng and Li-Chen Fu demonstrated a hybrid system of applying homeostatic drives theory and stimuli from environment or human on a robot, aiming to solve the increasing need of elder caring in modern aging society. Through reinforcement learning with human feedback, the robot is enabled to make appropriate decisions by reviewing its own motivation, environment stimuli and human inputs, allowing the robot to finish vague tasks, i.e., to serve, help or have a conversation with human on different situations. In our experiment, the human satisfaction rate could reach 94% and the robot could as well maintain a stable condition to perform long-term service. If the robot is to be deployed in a nursing home, it can further serve and meet the requirements of elders.

7. In Oct 2007, S. C. Mukhopadhyay and G. Sen Gupta proposed a development work of a system for elder care configured around sensors and robotic environment. A few sub-systems have been amalgamated to achieve the desired objective. The integrated system is able to support people who wish to live alone but, because of old age, ill health or disability, are at risk and are a cause of concern for their family and friends. The first part of the integrated system, named Selective Activity Monitoring (SAM) system, works on the principle of using sensor units (SU) to monitor various electrical appliances in a house. Rules, based on the daily activities

of a person, are defined for the appliances to be monitored. The system can detect violation of the rules and generate an alarm. The second part of the system is a low-cost Physiological Parameters Monitoring (PPM) system, called Medicmate, which can be used to monitor physiological parameters of a human subject such as body temperature, blood oxygen level and heart rate.

8. In June 2014, Bhuvaneswari.P.T.V, Vignesh.S, Papitha.S, Dharmarajan.R.S proposed an Orthopaedic problems have been a constant and nagging hindrance that has been plaguing the elderly community. In pursuit of finding a solution to this problem, a Physiotherapeutic Assistive Trainer (PAT) for elders using a Humanoid Robot, named Nao has been developed. Seamless interaction between Nao and the elderly patient is achieved using the Text-to-Speech module and sensors present in Nao. Further, it replaces human trainer by carefully monitoring the performance of the elders through the camera, equipped in Nao. The array of physiotherapeutic exercises executed by Nao is repeated when the patient makes their request by touching Nao's tactile sensor.

9. In Oct 2015, Yan Peng, Dong Qu , Yuxuan Zhong, Shaorong Xie, Jun Luo, Jason Gu Presented an Obstacle avoidance ability is the significant embodiment of the ground mobile robot, and the basic guarantee of the ground mobile robot to perform various tasks. Obstacle avoidance technologies are divided into two kinds, one is based on the global map and another is based on sensors respectively. This paper mainly aims at the local obstacle avoidance method based on sensors. The study of obstacle detection and obstacle avoidance are two inseparable parts in the research of obstacle avoidance ability. This paper proposes an efficient obstacle detection and obstacle avoidance algorithm based on 2-D lidar.

10. In the year 2019, T R Madhavan and M Adharsh proposed an Obstacle Detection and Obstacle Avoidance are vital technologies for autonomous navigation of vehicles. A number of sensor technologies have been used before to perform obstacle detection and obstacle avoidance. One kind of such sensor is a 2- D lidar sensor renowned for its accuracy in measuring distances. This research involves usage of low cost 2-D lidar system from Slamtech for obstacle detection and avoidance. This paper proposes an efficient algorithm for the implementation of obstacle detection and avoidance using 2-D lidar. A method is proposed as to how to get the data from the lidar sensor and also an algorithm is designed to get the information of obstacles by filtering, clustering technique. A systematic approach is used to avoid obstacles on the principle of minimum cost function. The efficacy of the algorithm implemented is verified using MATLAB simulation platform.

CHAPTER 3

METHODOLOGY

3.1 Artificial Intelligence in Elderly Care Robots

Artificial intelligence is a key component of the future of healthcare. Indeed, the era of the AI doctor seems to be unavoidable. Today, we see AI in hospitals helping clinicians identify medical risks; predict when to provide targeted, life-saving interventions; form treatment plans for patients with rare diseases; and deliver precision medicine. However, one of the benefits associated with AI may actually be a disadvantage in healthcare. While AI excels at making unbiased, purely logical decisions, it cannot yet appreciate the complex blend of emotional, social, cultural and physical needs of people. If left in the hands of a purely AI-driven doctor, the diagnosis and treatment plan may be correct, but it may not be what the patient will respond best to. Human doctors understand how to treat people, rather than just diseases. They will read the non-verbal clues, understand when to probe more deeply for a better reading of the patient's condition, and the type of care plan that is most likely to work for that individual. The human doctor builds a relationship with their patient, even if only fleeting, based on emotional connections. If the human touch is fundamental to healthcare, can AI ever step in to play a role in the softer side of care?

With massive improvements in healthcare and lifestyle in the last century, people are living longer. While longevity is an important achievement of modern-day, it does present challenges in terms of caring for an increasingly elderly population.

The issue of elder care is one of supply and demand. As the elder population increases in numbers, the number of caregivers is not correspondingly increasing. Japan predicts a shortage of 1 million caregivers by the year 2025. And the United States predicts that the percentage of people aged 65 and older is expected to increase

to roughly 26% of the population by 2050. The relative lack of people able to care for the elderly is what keeps the costs of elderly care so high and creates a burden on family and caregivers.

Using medical robots for elderly care will vastly reduce the current astronomical cost of elderly care. Additionally, it will pick up the slack in terms of the number of caregivers available as the ratio of elderly to nonelderly people shifts. Medical robots are a growing industry. According to the International Federation of Robotics World Robotics 2018 Service Robots report, medical robot sales increased 73% in 2017 over 2016, accounting for 2.7% of all professional service robot sales.

There are numerous ways that medical robotics can help the elderly:

- Robots can perform small tasks like fetching food and water.
- Some elder care robots handle social and emotional needs by providing entertainment through games, helping remind them of events and appointments, and providing social engagement.
- Other eldercare robots take a more direct, muscular approach and use powerful hydraulics to help provide mobility and transportation support to seniors.

3.2 Components of a Robot

A robot is a programmable mechanical device that can perform tasks and interact with its environment, without the aid of human interaction. Robotics is the science and technology behind the design, manufacturing and application of robots. The components of a robot are the body/frame, control system, manipulators, and drivetrain. Body/frame: The body or frame can be of any shape and size. Essentially, the body/frame provides the structure of the robot.

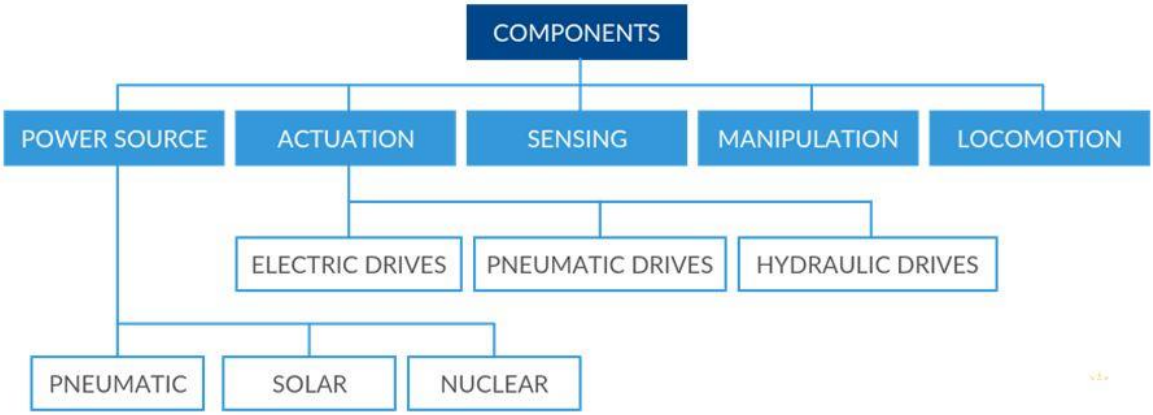

Fig. 3.1 Components of a Robot.

3.2.1 Body/Frame & Chassis:

The body or frame can be of any shape and size. Essentially, the body/frame provides the structure of the robot. Most people are comfortable with human-sized and shaped robots that they have seen in movies, but the majority of actual robots look nothing like humans. Typically, robots are designed more for function than appearance.

Fig. 3.2 Chassis of Robot

3.2.2 Control System:

The control system of a robot is equivalent to the central nervous system of a human. It coordinates and controls all aspects of the robot. **Sensors** provide feedback based on the robot's surroundings, which is then sent to the Central Processing Unit (CPU). The CPU filters this information through the robot's programming and makes decisions based on logic. The same can be done with a variety of inputs or human commands.

Fig. 3.3 CPU (Jetson Nano)

3.2.3 Manipulators:

To fulfil their purposes, many robots are required to interact with their environment, and the world around them. Sometimes they are required to move or reorient objects from their environments without direct contact by human operators. Unlike the Body/frame and the Control System, manipulators are not integral to a robot, i.e., a robot can exist without a manipulator.

Fig. 3.4 Manipulator of a Robot

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3.2.4 Electric motors (DC/AC):

Motors are electromechanical component used for converting electrical energy into its equivalent mechanical energy. In robot's motors are used for providing rotational movement.

Fig. 3.5 DC motor(12v)

3.2.5 Sensors:

Sensors provide real time information on the task environment. Robots are equipped with tactile sensor it imitates the mechanical properties of touch receptors of human fingerprints and a vision sensor is used for computing the depth in the environment.

3.2.6 Power Supply:

The working power to the robot is provided by batteries, hydraulic, solar power, or pneumatic power sources.

Table 3.1 List of Components

S.NO	COMPONENTS USED
1	NVIDIA JETSON NANO board (4gb)
2	STEREO Camera & Mic
3	LCD Display (5 inch)
4	DC Motors (12V), 10kg high torque and DC Motor Driver
5	Audio Amplifier and Speaker
6	Heart Rate Monitoring sensor
7	Chassis and wheel
8	SMPS or Battery

3.3 Voice assistant:

A typical conversational AI application uses three subsystems to do the steps of processing and transcribing the audio, understanding (deriving meaning) of the question asked, generating the response (text) and speaking the response back to the human. These steps are achieved by multiple deep learning solutions working together. First, automatic speech recognition (ASR) is used to process the raw audio signal and transcribing text from it. Second, natural language processing (NLP) is used to derive meaning from the transcribed text (ASR output). Last, speech synthesis or text-to-speech (TTS) is used for the artificial production of human speech from text. Optimizing this multi-step process is complicated, as each of these steps requires building and using one or more deep learning models. When developing a deep learning model to achieve the highest performance and accuracy for each of these areas, a developer will encounter several approaches and experiments that can vary by domain application.

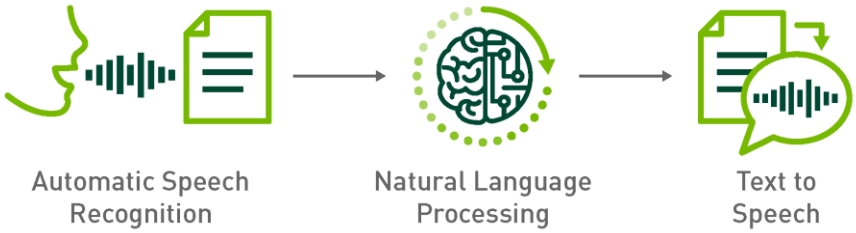

Fig 3.6 Process involved in Voice Assistant

3.3.1 Automatic Speech Recognition (ASR):

Automatic speech recognition refers to technologies built to process human speech and turn it into text. The first take at speech recognition dates back to 1952 when Three Bell Labs researchers built a system called “Audrey” for single-speaker

digit recognition. Advanced versions of ASR technologies now incorporate what is known as Natural Language Processing (NLP). These capture real conversations between people and use machine intelligence to process them. The accuracy provided by ASR is dependent on many factors, including speaker volume, background noise, the recording equipment used and more.

The process works as follows:

- An individual or a group speaks and an ASR software detects this speech.
- The device then creates a wave file of the words it hears.
- The wave file is cleaned to delete background noise and normalize the volume.
- This filtered wave form is then broken down and analyzed in sequences.
- The automatic speech recognition software analyzes these sequences and employs statistical probability to determine the whole words and then complete sentences.
- Some technology providers check the ASR's work with professional human transcribers who correct any errors to achieve greater accuracy.

3.3.2 Natural Language Processing (NLP):

The field of natural language processing (or NLP) is vast and consists of many subdomains or areas of study, such as document classification, machine translation, information extraction, information representation, summarization and language generation to name a few.

The two aspects of the larger set of NLP domains that are used in building voice assistants or even the tech that underlines chatbots are classification and information extraction or tagging. Classification in the context of conversational systems is to find the intent of the spoken sentence or loosely the action that has to be performed, and tagging is to find useful information from within the sentence which is needed to complete the task.

3.3.3 Speech Synthesis or Text-To-Speech (TTS):

Mean Text to speech technology refers to a field of computer science that enables the conversion of language text into audible speech. Also known as voice computing, text to speech (TTS) often involves building a database of recorded human speech to train a computer to produce sound waves that resemble the natural sound of a human speaking. This process is called speech synthesis.

Once a text to speech engine has generated the text it intends to convert to speech, it needs to produce the sounds required for articulation. This stage of the process involves converting language characters into phonemes, or distinct sounds. To achieve this, the text to speech engine must understand the context of the sentence in order to determine the proper tense.

3.4 Obstacle Avoidance using Machine Learning

Machine learning is an application of Artificial Intelligence (AI) that provides systems the ability to automatically learn and improve from experience without being explicitly programmed. Machine learning focuses on the development of computer programs that can access data and use it to learn for themselves.

In machine learning, tasks are generally classified into two broad categories. These categories are based on how learning is received or how feedback on the learning is given to the system developed.

- Unsupervised learning
- Supervised learning

3.4.1 Unsupervised learning

Unsupervised learning is the training of an AI algorithm using information that is neither classified nor labelled and allowing the algorithm to act on that information without guidance. In unsupervised learning, an AI system may group unsorted information according to similarities and differences even though there are

no categories provided. Unsupervised learning algorithms can perform more complex processing tasks than supervised learning systems. However,unsupervised learning can be more unpredictable than the alternate model. There are two types of unsupervised learning:

- Clustering
- Association

3.4.1.1 Clustering

Clustering is an important concept when it comes to unsupervised learning. It mainly deals with finding a structure or pattern in a collection of uncategorized data. Clustering algorithms process data and find natural clusters (groups) if they exist in the data. Some of the common clustering algorithms include:

- Hierarchical clustering
- K-means clustering
- K-NN (k nearest neighbors)
- Principal Component Analysis
- Singular Value Decomposition
- Independent Component Analysis

3.4.1.2 Association

Association learning is a rule based machine learning and data mining technique that finds important relations between variables or features in a data set. It identifies hidden correlations in databases by applying some measure of interestingness to generate an association rule for new searches.

Association rule algorithms count the frequency of complimentary occurrences, or associations, across a large collection of items or actions. The goal

is to find associations that take place together far more often than finding in a random sampling of possibilities. This rule-based approach is a fast and powerful tool for mining categorized, non-numeric database

3.4.2 Supervised learning

Supervised machine learning builds a model that makes predictions based on evidence in the presence of uncertainty. A supervised learning algorithm takes a known set of input data and known responses to the data (output) and trains a model to generate reasonable predictions for the response to new data. There are two types of supervised learning:

- Classification
- Regression

3.4.2.1 Classification

Classification techniques predict discrete responses. Classification models classify input data into categories. Typical applications include medical imaging, speech recognition, and credit scoring. Classification is used to tag, categorize or separate data into specific groups or classes. For example, applications for hand writing recognition use classification to recognize letters and numbers. Common algorithms for performing classification include

- Discriminant analysis
- Boosted and bagged decision trees
- K-nearest neighbor
- Support vector machine (SVM)

3.4.2.2 Regression

Regression techniques predict continuous responses. Typical applications include electricity load forecasting and algorithmic trading. Regression technique

is used if working with a data range or if the nature of your response is a real number, such as temperature or the time until failure for a piece of equipment. Common regression algorithms include:

- Linear regression
- Ensemble methods
- Decision trees
- Neural Networks

3.5 Classification algorithms

3.5.1 Discriminant analysis

Discriminant analysis is a classification method which assumes that different classes generate data based on different Gaussian distributions. It is a generalization of Fisher's linear discriminant, a method used in statistics, pattern recognition, and machine learning to find a linear combination of features that characterizes or separates two or more classes of objects or events. The resulting combination may be used as a linear classifier or, more commonly, for dimensionality reduction before later classification.

3.5.2 Boosted and bagged decision trees

Decision trees are used to create training models which can be used to predict class or value of target variables by learning decision rules inferred from training data. Unlike other supervised learning algorithms, decision tree algorithmscan be used for solving both regression and classification problems. The decision tree algorithm tries to solve the problem, by using tree representation. Each internal node of the tree corresponds to an attribute, and each leaf node corresponds to a class label. Decision tree is simple to understand and visualize, requires little data preparation and can handle both numerical and categorical data but it can create complex trees that do not generalize well. Decision trees can be unstable because small variations in the data might result in completely different tree being generated.

3.5.3 K-nearest neighbor

K-nearest neighbor is a data classification algorithm that attempts to determine what group a data point is by looking at the data points around it. The k-nearest neighbor is an example of a "lazy learner" algorithm because it does not attempt to construct a general internal model, but simply stores instances of the training data. Classification is computed from a simple majority vote of the k nearest neighbors of each point. The only calculations it makes are when it is askedto poll the data point's neighbors. This makes k-nn very easy to implement for data mining. This algorithm is simple to implement, robust to noisy training data, and effective if training data is large. The main disadvantage of k-nn is the need to determine the value of K. The computation cost is high as it needs to compute the distance of each instance to all the training samples.

3.5.4 Support vector machine

Support vector machines are supervised learning models with associated learning algorithms that analyze data used for classification and regression analysis. Given a set of training examples, each marked as belonging to one or the other of two categories, an SVM training algorithm builds a model that assigns new examples to one category or the other, making it a non-probabilistic binary linear classifier. The support-vector clustering algorithm applies the statistics of support vectors (data points that are closer to the hyperplane and influence the position and orientation of the hyperplane), developed in the support vector machines algorithm, to categorize unlabelled data. This is one of the most widely used clustering algorithms in industrial applications. It is effective in high dimensional spaces and uses a subset of training points in the decision function soit is also memory efficient.

3.6 Deep Learning

Deep Learning is an artificial intelligence function that imitates the workings of the human brain in processing data and creating patterns for use in decision making. Deep learning is a subset of machine learning that has networks capable of learning unsupervised from data that is unstructured or unlabelled.

Deep learning utilizes a hierarchical level of artificial neural networks to carry out the process of machine learning. The artificial neural networks are built like the human brain, with neuron nodes connected together like a web. While traditional programs build analysis with data in a linear way, the hierarchical function of deep learning systems enables machines to process data with a nonlinear approach.

In Deep Learning networks, each layer of nodes trains on a distinct set of features based on the previous layer's output. The further you advance into the neural net, the more complex the features your nodes can recognize, since they aggregate and recombine features from the previous layer. Deep Learning networks are distinguished from the single-hidden-layer neural networks by their depth, i.e. the number of node layers through which data must pass in a multistep process of pattern recognition.

Deep Learning then can be defined as neural networks with a large number of parameters and layers in one of four fundamental network architectures:

- Convolutional Neural Network
- Recurrent Neural Network
- Recursive Neural Network
- Pretrained Network

3.6.1 Convolutional Neural Network

Convolutional Neural Networks (CNN) is a type of deep neural network architecture designed for specific tasks like image classification. CNNs were inspired by the organization of neurons in the visual cortex of the animal brain. A CNN is composed of an input layer. For basic image processing, this input is a two-dimensional array of neurons which correspond to the pixels of an image. It also contains an output layer which is a one-dimensional set of output neurons. CNN uses a combination of sparsely connected convolutional layers, which perform image processing on their inputs. In addition, they contain downsampling layers called pooling layers to further reduce the number of neurons necessary in subsequent layers of the network. And finally, CNNs contain one or more fully connected layers to connect the pooling layer to the output layer. CNNs work well for a variety of tasks including image recognition, image processing, imagesegmentation, video analysis, and natural language processing.

3.6.2 Recurrent Neural Network

The Recurrent Neural Network (RNN) can operate effectively on sequences of data with variable input length. This means that RNN uses knowledge of its previous state as an input for its current prediction, and this process can be repeated for an arbitrary number of steps allowing the network to propagate information via its hidden state through time. This feature makes RNNs very effective for working with sequences of data that occur over time. RNNs work wellfor applications that involve a sequence of data that change over time. These applications include natural language processing, speech recognition, language translation, image captioning and conversation modelling.

3.6.3 Recursive Neural Network

A recursive neural network is a kind of deep neural network created by applying the same set of weights recursively over a structured input, to produce a structured prediction over variable-size input structures or a scalar prediction on it by traversing a given structure in topological order. Recursive Neural Networks have been successful in learning sequence and tree structures in natural language processing, mainly phrase and sentence continuous representations based on word embedding.

3.6.4 Pretrained Network

A pretrained network is a model that was trained on a large benchmark dataset to solve a problem similar to the one that we want to solve. Accordingly, due to the computational cost of training such models, it is common practice toimport and use such pre-trained networks (e.g. Alexnet, squeezenet, googlenet). The process of importing and using a pre-trained network is referred to as transfer learning. Transfer learning is a popular method in computer vision because itallows us to build accurate models in a timesaving way. With transfer learning, instead of starting the learning process from scratch, it can be started from patterns that have been learned when solving a different problem.

3.7 Transfer Learning

The pretrained network should initially be loaded to the workspace. Thefinal layers of the network (fully connected layer and classification layer) must be replaced according to the problem at hand. Once the final layers are replaced, the network has to be trained for a certain number of epochs. A certain number of iterations take place within a single epoch. For every iteration, the weights allocated to each convolutional layer are adjusted to reduce the error in prediction. The

accuracy of the trained network is then predicted by determining the number of correct predictions the network makes. The network is then saved and tested andthe accuracy can be further improved by retraining the saved network.

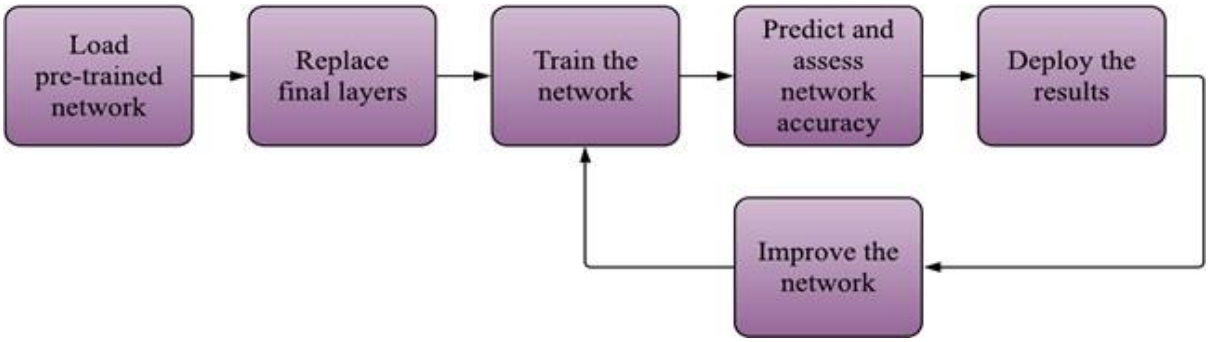

Fig 3.7 Steps to reuse pretrained network

CHAPTER 4

AI Based Interactive Voice Assistant

4.1 Introduction

In this project for voice assistant we have used python programing language to build the voice assistant and we have named the voice assistant as 'TED'.

4.2 Automatic Speech Recognition(ASR)

The microphone is used to get the input voice of the people. This is then converted to text and then it checks for the trigger key 'TED' and if the trigger key is present in the input voice. Then the voice assistant gets active.

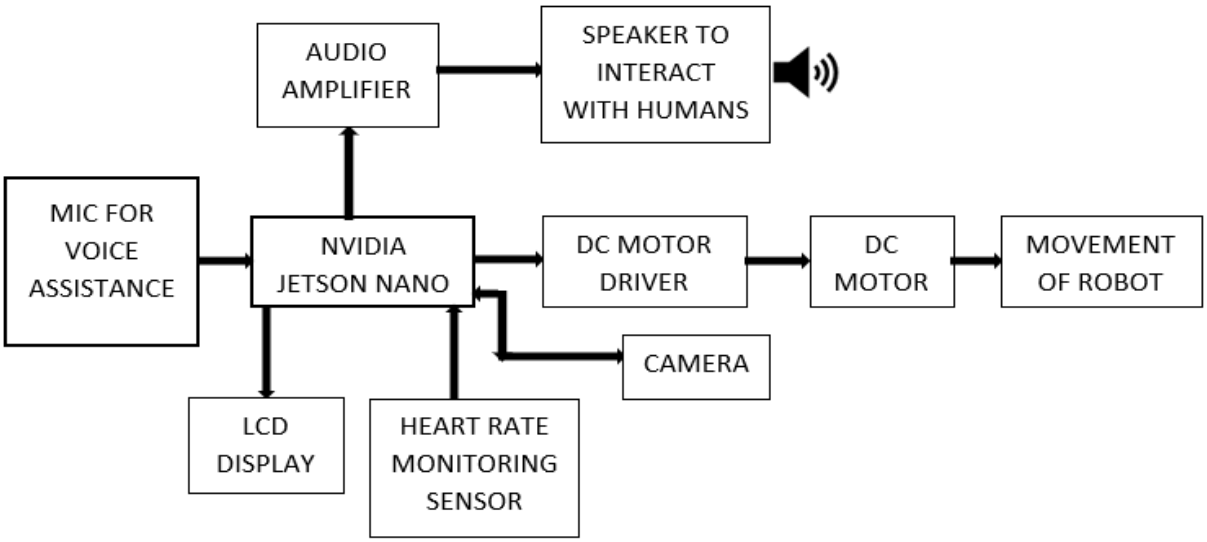

Fig 4.1 Block Diagram of the Robot.

4.3 Natural Language Processing (NLP)

The fetched data from the analog voice input is then processed with respect to the given datasets and scenarios and compares with the output that has to be displayed if the reply is satisfied then it goes for the next step.

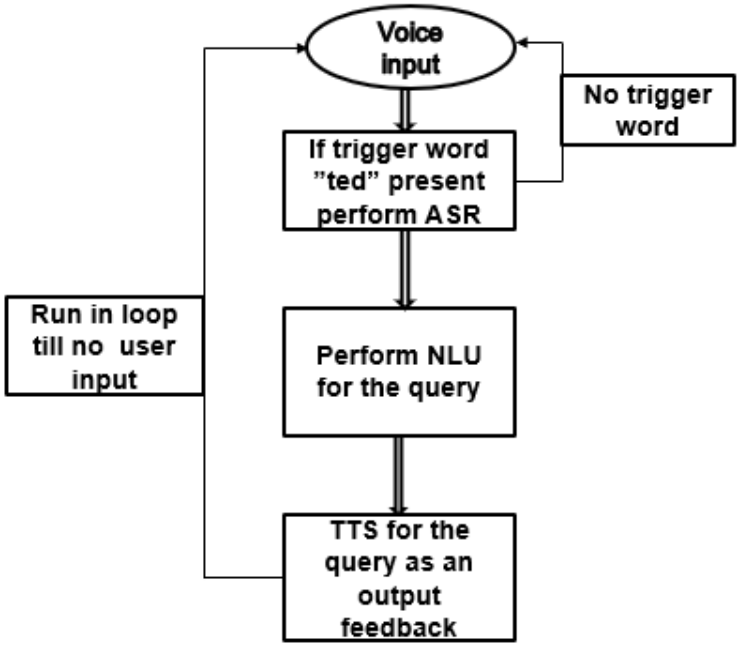

Fig 4.2 working of voice assistant

4.4 Speech Synthesis or Text-To-Speech (TTS)

The Data is then speech synthesized and converted to text-to-speech and gives feedback in the form of audio signal to the speakers. By using this method we can make the robot to get interacted with the people.

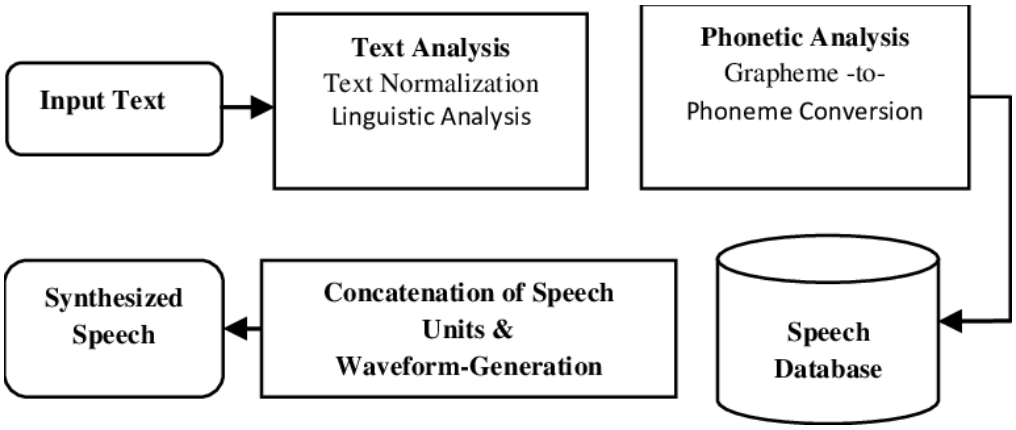

Fig 4.3 Process of Text-to-Speech

4.5 Results

By following these three methods we created a voice assistant that will interact with the elderly people under any situation and scenarios.

CHAPTER 5

Obstacle Avoidance

5.1 Image Processing using STEREO CAMERA

Robust navigation for mobile robots over long distances requires an accurate method for tracking the robot position in an unstructured environment. Generally, position information is obtained from wheel encoder, inertial sensors, GPS and range sensors. Wheel encoder is often deceived by slippage, inertial sensors are prone to internal drifts and GPS signal is poor in areas like forests, inside mines or while operating on a planet that has not yet been equipped with a set of GPS satellites. The stereo based navigation is more promising considering the limitations of existing systems, e.g., Spirit and Opportunity Rovers exploration on Mars. Moreover, planet exploration missions demand autonomous navigation where large communication delays are involved (in terms minutes) and environment is harsh. Lately, Stereo vision has become highly demanding research field due to the increase of high computational power and reduction in cost. Recent applications of stereo vision on different types of robots in various environments for navigation and mapping can be founded.

5.2 Training of the Model

Stereo vision system is composed of two cameras. with the aim of obtaining 3D geometric information of the robot's surroundings and pose estimation of a robot. The obstacles avoidance by potential field method is applied along with navigation control loop for mobile robot.

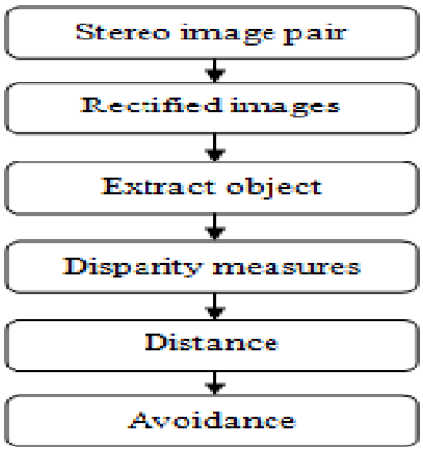

Fig 5.1 Process involved in Stereo camera image processing

This flowchart shows the process involved in the navigation of the robot using stereo camera. First of all, the stereo camera pairs the images ad then rectifies the images with the previous images. Then it extracts only the objects from the images and then measures the depth and distance of the obstacles from the robot. Then it avoids the obstacles and changes the direction.

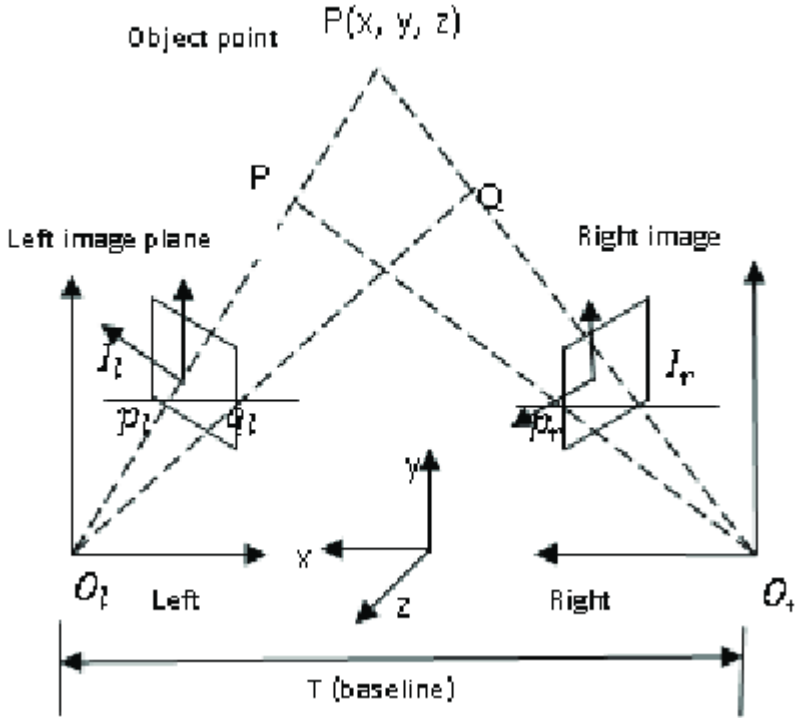

Fig 5.2 Geometry of Stereo Camera

The above diagram shows the geometry of the stereo camera and the angle by which it can capture the objects in front of it and avoids the obstacles. It captures two images from left and right side.

5.3 Result

Real-time obstacle detection and avoidance is one of the key issues in autonomous navigation of mobile robots. All the autonomous robots incorporate some kind of collision avoidance, ranging from primitive algorithms that detect an obstacle and guide the robot in order to avoid collision through sophisticated algorithms. Stereo vision sensors, which provide dense depth maps help in resolving small or narrow objects significantly better than other sensors like sonar sensors, Laser sensor etc. Laser finds high degree of accuracy but takes longer time to scan the area. Hence, with the aim of acquiring more accurate information about size, shape and position of the obstacle in the environment, stereo camera based system is proposed in this project.

CHAPTER 6

APPLICATION DEVELOPMENT

6.1 Purpose

A **graphical user interface** (**GUI**) is a type of user interface through which users interact with electronic devices via visual indicator representations. To have a better interaction with the robot we have created a GUI for the user to get the feedback from the robot.

6.2 Creation of Graphical User Interface

We have created a GUI for the interaction with the elderly people and that will be simple to get understand and easy to interact.

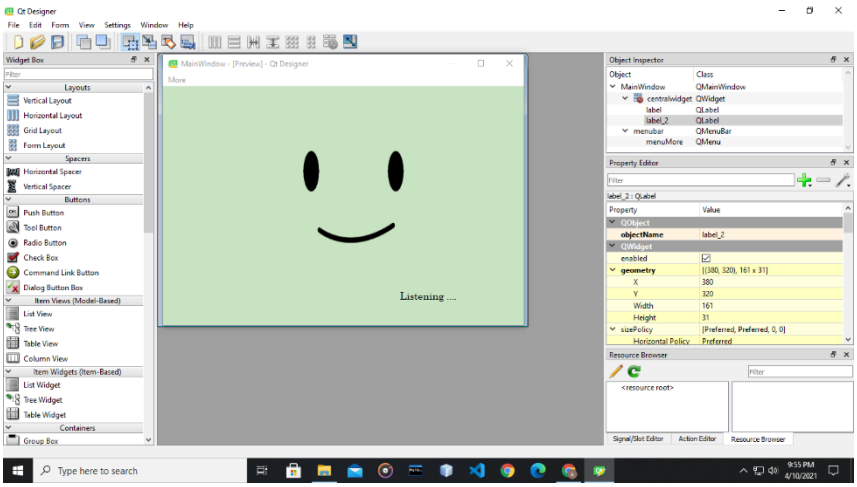

Fig. 6.1 Creating GUI for user interface

In this GUI we have included an emoji like facial expression for the robot to express its way of conversation. And we have made the details which will be easily understood by the users and we can set remainders and also feed data to the robot from the GUI.

6.3 Working of GUI

After adding the facial expression, we added some basic information like time, date, remainders and trending news. In this application we can easily understand the basic buttons and menu present in it. For every button we have coded what should be the action to be performed at the backend. And to look neat and easy to work we have used several multimedia tools for developing the application. The uses of a pointer that serves as navigation to interact with different visually appealing Graphical icons. Abstraction is a major concept that has been used in a GUI operating system. Users can use the pointer to click on the icon, which initiates a series of actions. Normally an application or functionality will get started. Then the user will have to provide input or tasks to generate the desired action from the machine. The GUI actually translates user language, which comprises simple one-line commands, single click and double clicks to machine language or assembly language. The machine understands machine language, and hence the machine responds to the task initiated, which is translated to use language and communicated to the user via GUI.

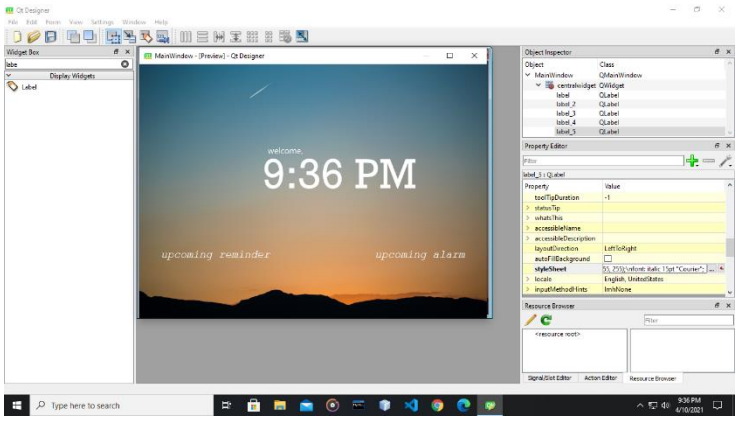

Fig 6.2 Frontend of GUI

There are many benefits to using a GUI. Apart from the fact that it provides users with an intuitive and easy-to-use interface and immediate visual feedback, a GUI also allows a user to open up multiple programs or instances and displays these simultaneously.

CHAPTER 7

CONCLUSION

In this project ‘AI Based Interactive Robot for Elderly People’ we have successfully made a Voice Assistant by using Artificial Intelligence for interacting with the elderly people and also to achieved the obstacle avoidance of the robot for its locomotion.

We have done a lot of R&D in building up the robot and the components that has to be used and we built the overall structure of the prototype and we have developed a voice assistant named as “TED” and this AI will interact with the elderly people and we have made it using python programming language and this interacts and responds with the elderly people. We also worked on the obstacle avoidance of the robot and we have achieved half way for the obstacle avoidance of the robot and we have created a GUI for user interface and to interact with the user.

Still few more features are needed to be added to make the robot be useful in all ways for the elderly people and this will be done in the future and still a lot of R&D has to be done to improve the features and its efficiency. The stereo camera helps in obstacle avoidance and by using this image processing technique we can make the robot to move on its own. These are all the overall work done and the progress of the robot.

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